

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	S.SRI MATHI	Reg. No.:	192424095
Section / Batch:	SLOT B	Date:	24.08.2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

Question Type	Focus Area	Questions
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1

SECTION A – Industry Problem Solving Task

Code of Conduct

I S.SRI MATHI (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION**Industry Problem Solving Task**

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%				
Application of Engineering Concepts	25%				

Solution Design	25%				
Use of Tools	15%				
Analysis	10%				
Professionalism	5%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

ASSESSMENT TOOL 2 – Industry Problem Solving Task

NAME: S.SRI MATHI

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COURSE CODE: CSA0603

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHM

DATE: 24.08.2026

Q35. Autonomous Drone Delivery Routing (Graph + Optimization)

A logistics company uses drones to deliver packages efficiently across a city. Analyze how path planning algorithms can optimize delivery routes. Identify constraints such as battery life and obstacles. Design a solution using graph-based optimization techniques. Discuss scalability for multiple drones. Evaluate performance metrics. Interpret results in logistics automation.

1. Problem Statement

A logistics company wants to use **autonomous drones** to deliver packages from a warehouse or drone station to multiple customer locations.

The main problem is to find a route that is:

- **Shortest or fastest**
- Safe from obstacles and restricted areas
- Within the drone's **battery range**
- Able to carry the required package weight
- Efficient when several drones are operating simultaneously

This is a real-world **graph optimization problem**. The delivery locations and possible flight paths can be represented as a graph, and optimization algorithms can be used to select efficient routes.

2. Real-Time Industry Scenario

This technology is mainly used in the **logistics, e-commerce, food delivery and healthcare industries**.

A strong current example is **Amazon Prime Air**. As of August 2026, Amazon announced plans to expand drone delivery to nearly **500 U.S. cities and towns by the end of 2026**. Its MK30 drone can operate up to about **7.5 miles from a delivery facility**, and Amazon uses onboard cameras, sensors and a Detect-and-Avoid system for autonomous navigation.

Another current example is **DoorDash Air**, which received FAA Part 135 certification in July 2026 and is developing its own drone delivery operation. DoorDash is integrating drones with its existing delivery network rather than treating drones as a completely separate system.

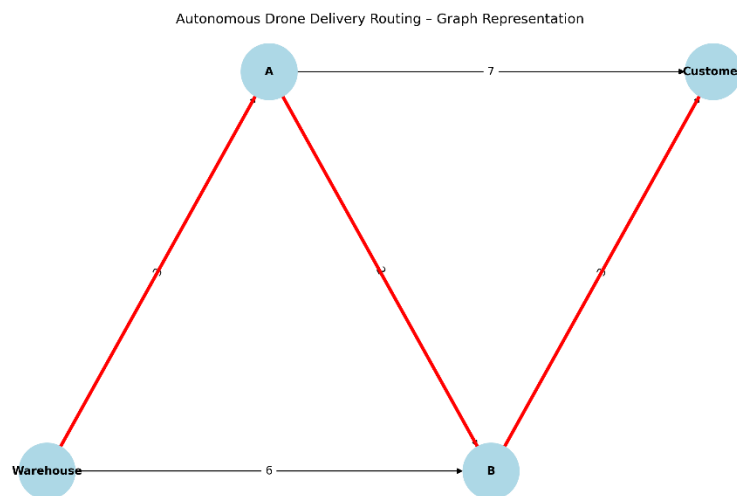
Also, in August 2026, **Uber and Zipline announced a partnership** to introduce autonomous drone delivery for Uber Eats in the U.S., showing that drone routing is becoming part of mainstream food-delivery logistics.

So this problem is not theoretical. **Autonomous route optimization is an important part of modern last-mile logistics.**

3. Graph Representation

The city can be represented as a **weighted graph**:

$$G=(V,E)$$



Where:

- **V (Vertices)** = warehouse, drone stations, customer locations, charging stations and safe landing/drop points.

- **E (Edges)** = possible flight paths between locations.
- **Edge weight** = distance, travel time or energy required.

4. Constraints in Drone Routing

A simple shortest-path problem is not enough because a real drone has physical and operational limitations.

a) Battery Constraint

The drone cannot travel farther than its available battery allows.

For example:

$$\text{Energy}(\text{Route}) \leq \text{BatteryCapacity}$$

If a route requires 80% battery but the drone has only 70% available, that route must be rejected.

The system should also maintain a **safety reserve** instead of allowing the battery to reach zero.

b) Obstacles

The drone must avoid:

- Buildings
- Towers
- Bridges
- Power lines
- Trees
- Restricted airspace
- Other aircraft/drones

An obstacle can be represented by removing an edge from the graph or assigning it a very high cost.

Amazon's current Prime Air system, for example, uses onboard sensing and a Detect-and-Avoid system to make real-time decisions about its surroundings.

c) Payload Constraint

The drone has a maximum carrying capacity.

$\text{PackageWeight} \leq \text{DroneCapacity}$

A package exceeding the drone's capacity cannot be assigned to that drone.

d) Delivery Time

Some packages may have a deadline.

For example:

Food order must reach the customer within 30 minutes.

Therefore:

$\text{DeliveryTime} \leq \text{Deadline}$

e) Weather

Rain, strong wind and poor visibility can affect:

- Battery consumption
- Flight speed
- Safety
- Route availability

Therefore, the routing system should dynamically update the graph when weather conditions change.

5. Optimization Approach

A suitable solution can combine **Greedy algorithms, Dynamic Programming and graph algorithms.**

Step 1 – Construct the Graph

Create a graph containing:

- Warehouse
- Customers
- Charging stations
- Safe flight corridors
- Obstacles

Assign weights such as:

Weight=Distance+EnergyCost+RiskCost

The lowest-weight route becomes preferable.

6. Greedy Strategy – Dijkstra's Algorithm

For finding the shortest path between two locations, **Dijkstra's algorithm** can be used when edge costs are non-negative.

Working

1. Start from the warehouse.
2. Set warehouse distance to 0.
3. Set all other nodes to infinity.
4. Select the unvisited node with the smallest cost.
5. Update the costs of its neighboring nodes.
6. Repeat until the destination is reached.

Example

If:

Warehouse → A = 3 km

Warehouse → B = 6 km

A → B = 2 km

B → Customer = 3 km

A → Customer = 7 km

The algorithm compares:

Warehouse $\rightarrow$ A $\rightarrow$ Customer
= 3 + 7
= 10 km

and

Warehouse $\rightarrow$ A $\rightarrow$ B $\rightarrow$ Customer
= 3 + 2 + 3
= 8 km

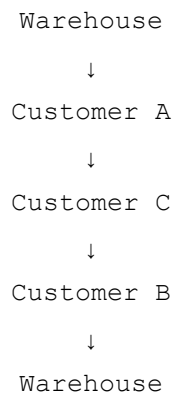
Therefore:

Selected route = Warehouse $\rightarrow$ A $\rightarrow$ B $\rightarrow$ Customer

7. Dynamic Programming for Delivery Planning

When the drone has to visit **multiple customers**, the problem becomes more complex.

For example:



The system must determine the best order of visiting customers.

This resembles the **Travelling Salesman Problem (TSP)** and, with multiple drones, the **Vehicle Routing Problem (VRP)**.

Dynamic programming can be used for smaller numbers of delivery locations by storing solutions to previously solved subproblems.

For example:

DP[S][i]

can represent the minimum cost of visiting the set of customers S and ending at customer i.

This avoids repeatedly calculating the same routes.

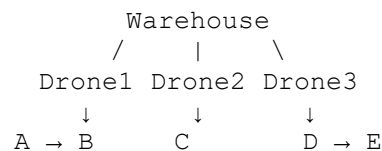
Important limitation

Dynamic programming becomes expensive as the number of customers increases.

Therefore, it is suitable mainly for **small or moderate routing problems**, while large real-world systems generally need scalable heuristics, optimization methods and real-time planning.

8. Multi-Drone Optimization

Suppose there are:



The problem is no longer just:

"What is the shortest route?"

It becomes:

Which drone should deliver which package, and what route should each drone follow?

This is a **Multi-Drone Vehicle Routing Problem**.

The system should optimize:

- Drone assignment
- Delivery sequence
- Total distance
- Battery usage
- Delivery time

- Collision avoidance
- Charging requirements

A practical system can use:

Greedy algorithm → initial assignment

followed by:

Optimization/DP/heuristic → improve routes

And **Real-time replanning → respond to obstacles/weather/battery changes**

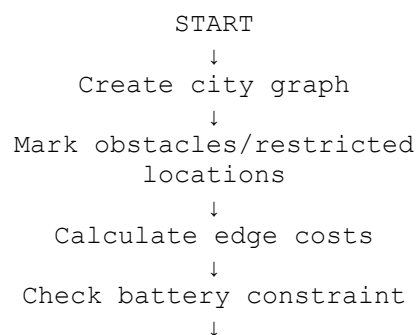
Recent research in 2026 is also exploring multi-robot task allocation and routing using optimization methods such as ant colony optimization. One recent study reported improvements in travel distance and completion time compared with baseline methods.

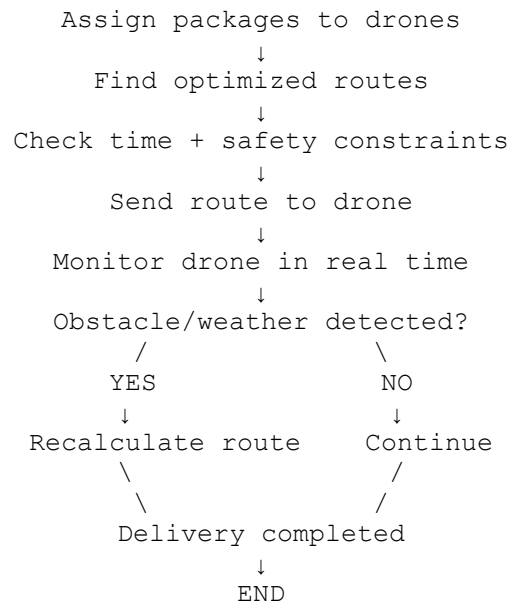
9. Proposed Algorithm

Input

- Number of drones
- Customer locations
- Warehouse location
- Distances
- Battery capacities
- Package weights
- Obstacles
- Delivery deadlines
- Weather conditions

Processing





10. Pseudocode

START

Input:

Graph G
Warehouse W
Customers C
Drones D
Battery capacity
Package weights
Obstacles

For each drone:
 Check available battery
 Check package weight

Remove unsafe/blocked edges from G

For each customer:
 Find shortest feasible route
 using graph-based optimization

 If battery constraint is violated:
 Find charging station
 Recalculate route

Assign customers to drones
while minimizing:
 total distance
 energy consumption
 delivery time

For multiple customers:
 optimize delivery sequence

During flight:
 monitor battery

```
monitor obstacles
monitor weather
```

```
If conditions change:
    update graph
    recalculate route
```

```
Return optimized delivery plan
```

```
END
```

11. Performance Evaluation

The proposed system can be evaluated using the following metrics:

Metric	Purpose
Total distance	Measures route efficiency
Delivery time	Measures speed
Battery consumption	Measures energy efficiency
Number of successful deliveries	Measures reliability
Route computation time	Measures algorithm speed
Obstacle avoidance rate	Measures safety
Battery-return failures	Measures operational reliability
Drone utilization	Measures fleet efficiency
Cost per delivery	Measures economic efficiency

12. Scalability

For **one drone and a few customers**, Dijkstra or dynamic programming can work effectively.

But as the number of drones and customers increases, the problem becomes much more complex.

Small system

```
1 Drone
5 Customers
    ↓
DP / Dijkstra
```

Medium system

```
5 Drones
50 Customers
    ↓
Greedy + optimization
```

Large system

100+ Drones
1000s of Customers
↓
Distributed optimization
+ heuristics
+ real-time replanning
+ AI/ML

A large-scale system should therefore avoid recalculating every possible route from scratch.

Instead, it can use:

- Hierarchical routing
- Greedy assignment
- Heuristic optimization
- Parallel computation
- Real-time graph updates
- Dynamic task allocation

This is particularly important because real delivery networks continuously change.

13. Real-Time Dynamic Routing

A major difference between a classroom graph and a real drone system is that **the graph changes while the drone is flying**.

For example:

Original:
Warehouse → A → B → Customer
↓
shortest

New obstacle detected:
Warehouse → A → ⓪ → Customer
↓
Recalculate
↓
Warehouse → C → D → Customer

Therefore, the drone should not blindly follow a route calculated before takeoff.

It should continuously receive information from:

- GPS
- Cameras
- LiDAR/sensors
- Weather systems
- Battery sensors
- Airspace information

and update the route when necessary.

Amazon's current Prime Air system explicitly uses onboard cameras and sensors for navigation and obstacle detection, along with its Detect-and-Avoid system.

14. Industry Interpretation

This solution has significant applications in **last-mile logistics**.

E-commerce

Companies can deliver small packages directly from fulfillment centers.

Food delivery

Restaurants can use drones for short-distance food deliveries, reducing dependence on road traffic.

Healthcare

Drones can transport:

- Medicines
- Blood samples
- Vaccines
- Emergency medical supplies

This is particularly useful where road transportation is slow or difficult.

Disaster management

Drones can deliver essential supplies to areas affected by:

- Floods
- Earthquakes
- Cyclones
- Landslides

Why it matters

Traditional delivery vehicles are affected by:

- Traffic
- Road conditions
- Fuel costs
- Driver availability

Drones can potentially bypass road congestion and provide faster point-to-point delivery, provided that battery, safety, airspace and weather constraints are satisfied.

15. Latest Industry Connection – 2026

The industry is moving from **experimental drone delivery toward integrated autonomous logistics networks.**

As of August 2026:

- **Amazon** is planning to expand Prime Air to nearly **500 U.S. cities and towns** by the end of 2026.
- Amazon's FAA documents describe MK30 operations with an operating range of about **7.5 miles** and proposed operations of up to **1,000 delivery flights per day from a delivery center**, depending on the specific approved operation.
- **DoorDash** has launched DoorDash Air after receiving FAA Part 135 certification, integrating drones with its broader autonomous delivery strategy.
- **Uber and Zipline** announced an autonomous drone-delivery partnership for Uber Eats, showing the movement toward combining drones with existing delivery platforms.

This proves why graph optimization is relevant: **when the number of drones, customers and delivery locations increases, manually planning routes becomes impossible.**

16. Advantages

1. Faster delivery

Drones can avoid road traffic.

2. Lower travel distance

Graph optimization selects efficient paths.

3. Better battery utilization

Energy constraints prevent inefficient routes.

4. Improved safety

Obstacle-aware routing avoids dangerous regions.

5. Better fleet utilization

Multiple drones can be assigned different customers.

6. Real-time adaptability

Routes can be recalculated when conditions change.

7. Reduced human dependency

Autonomous routing reduces manual route planning.

17. Limitations

The solution is not perfect.

Battery limitation

Current batteries restrict flight range and payload.

Weather

Strong wind, heavy rain and poor visibility can affect operations.

Airspace regulations

Drones cannot simply fly anywhere.

Computational complexity

Multi-drone routing becomes extremely difficult as the number of customers increases.

Infrastructure

Charging stations and drone hubs are required.

Safety and privacy

Autonomous drones operate near homes, roads and people.

Noise

Large-scale drone operations can create community concerns.

These are real deployment issues, not merely theoretical constraints. Current Amazon expansion plans, for example, still involve regulatory, environmental and community considerations.

18. Expected Outcome

The proposed graph-based optimization system should produce:

The safest and most efficient feasible route for each drone while minimizing distance, energy consumption and delivery time.

For multiple drones, the system should additionally minimize:

$$\text{Total Cost} = \text{Distance} + \text{Energy} + \text{Time} + \text{Risk}$$

subject to:

$$\text{Battery} \leq \text{Capacity} \quad \text{Payload} \leq \text{MaximumPayload} \quad \text{DeliveryTime} \leq \text{Deadline}$$

19. Conclusion

Autonomous drone delivery routing is a real-world optimization problem in the logistics and last-mile delivery industry. A city can be represented as a weighted graph where locations are vertices and possible flight paths are edges. Greedy graph algorithms such as Dijkstra's algorithm can identify efficient paths, while dynamic programming can help optimize delivery sequences for smaller routing problems.

For multiple drones, the problem becomes a multi-drone vehicle-routing and task-assignment problem. Battery capacity, payload, obstacles, weather, delivery deadlines and restricted airspace must be incorporated into the optimization process. Real-time monitoring allows the system to recalculate routes when conditions change.

The approach can reduce **delivery time, travel distance and battery consumption while improving drone utilization and operational safety.** The growing deployment of Prime Air, DoorDash Air and Uber-Zipline demonstrates that autonomous delivery is becoming an increasingly important part of modern logistics.